

Core Concept of Cryptography

Cryptography: is the science of keeping information secure, private, and authentic. it turns readable data (plaintext) into an unreadable scramble (ciphertext) so that only people with the right "key" can read it. Example: Think of it like a digital locked box: anyone can see the box exists, but you need the exact key to see what's inside.

The four pillars that make cryptography work:

1. Confidentiality: Ensuring only authorized people can read the message.
2. Integrity: Ensuring the message wasn't altered or messed with in transit.
3. Authentication: Proving the sender is actually who they claim to be.
4. Non-repudiation: Ensuring the sender cannot deny sending the message later.

There are two main types of cryptography:

1. Symmetric Cryptography (Secret Key) In symmetric cryptography, the exact same key is used to both lock (encrypt) and unlock (decrypt) the message. It is incredibly fast, but it has one massive flaw: both parties have to safely share the secret key beforehand. If an attacker steals the key in transit, the whole system collapses. Real-world example: AES (Advanced Encryption Standard), which secures everything from your Wi-Fi password to top-secret government files.
2. Asymmetric Cryptography (Public Key) To solve the key-sharing problem, asymmetric cryptography uses a matched pair of keys: Public Key: Anyone can see this. People use it to lock messages sent to you. Private Key: Only you have this. It is the only key that can unlock those messages. Real-world example: RSA or ECC (Elliptic Curve Cryptography). When you visit a website with <https://> and a padlock icon, your browser is using public-key cryptography to safely share a symmetric session key. The Hybrid Reality: Because symmetric encryption is fast and asymmetric encryption is secure, modern systems use both. Asymmetric encryption is used to safely exchange a temporary symmetric key, and then symmetric encryption handles the heavy lifting of scrambling the actual data.

Essential Foundations

The Essential Foundations To move into modern cryptography, you need a few toolkits in your belt. Don't worry, you don't need a PhD in math, just the basics: _ Binary & Bytes: How data is represented in computers (1s and 0s). _ The XOR Operator ($\oplus$): The holy grail of crypto programming. It's a simple logical operation, but it has perfect secrecy properties. _ Modular Arithmetic: Often called "clock math." If it's 10:00 and you add 4 hours, it's 2:00, not 14:00. This is the math that powers almost all modern encryptions.

- The extended Euclidean algorithm is an efficient way to find integers u, v such that $au + bv = \gcd(a, b)$
- We say that two integers are congruent modulo m if $a \equiv b \pmod{m}$ meaning a divided by m have remainder b .
- Perfect encryption: the one-time pad Essentially, a classical cipher can't be secure unless it comes with a huge key, but encrypting with a huge key is impractical. However, the one-time pad is such a cipher, and it is the most secure cipher. In fact, it guarantees perfect secrecy: even if an attacker has unlimited computing power, it's impossible to learn anything about the plaintext except for its length. In the next sections, I'll show you how a one-time pad works and then offer a sketch of its security proof. Encrypting with the One-Time Pad The one-time pad takes a plaintext, P , and a random key, K , that's the same length as P and produces a ciphertext C , defined as $C = P \oplus K$ where C , P , and K are bit strings of the same length and where $\oplus$ is the bitwise exclusive OR operation (XOR), defined as $0 \oplus 0 = 0$, $0 \oplus 1 = 1$, $1 \oplus 0 = 1$, $1 \oplus 1 = 0$

Classical Era of Cryptography

Before computers, cryptography was done with paper and pen. Starting here helps you understand the core concepts of encryption and decryption without the overwhelming math.

Classical cipher: _ The Caesar Cipher : The absolute basics. Shifting letters of the alphabet (e.g., $A \rightarrow D$). _ The Vigenère Cipher : The Vigenère cipher is similar to the Caesar cipher, except that letters aren't shifted by three places but rather by values defined by a key

- How Cipher work Based on simplistic ciphers like the Caesar and Vigenère ciphers, we can try to abstract out the workings of a cipher, first by identifying its two main components: a permutation and a mode of operation. A permutation is a function that transforms an item (in cryptography, a letter or a group of bits) such that each item has a unique inverse (for example, the Caesar cipher's three-letter shift). A mode of operation is an algorithm that uses a permutation to process messages of arbitrary size. The mode of the Caesar cipher is trivial: it just repeats the same permutation for each letter, but as you've seen, the Vigenère cipher has a more complex mode, where letters at different positions undergo different permutations.
- Not every permutation is secure. In order to be secure, a cipher's permutation should satisfy three criteria: • The permutation should be determined by the key, so as to keep the permutation secret as long as the key is secret. In the Vigenère cipher, if you don't know the key, you don't know which of the 26 permutations was used; hence, you can't easily decrypt. • Different keys should result in different permutations. Otherwise, it becomes easier to decrypt without the key: if different keys result in identical permutations, that means there are fewer distinct keys than distinct permutations, and therefore fewer possibilities to try when decrypting without the key. In the Vigenère cipher, each letter from the key determines a substitution; there are 26 distinct letters, and as many distinct permutations. • The permutation should look random, loosely speaking. There should be no pattern in the ciphertext after performing a permutation, because patterns make a permutation predictable for an attacker, and therefore less secure. For example, the Vigenère cipher's substitution is pretty predictable: if you determine that A encrypts to F, you could conclude that the shift value is 5 and you would also know that B encrypts to G, that C encrypts to H, and so on. However, with a randomly chosen permutation, knowing that A encrypts to F would only tell you that B does not encrypt to F.
- Why Classical Ciphers Are Insecure Classical ciphers are doomed to be insecure because they're limited to operations you can do in your head or on a piece of paper. They lack the computational power of a computer and are easily broken by simple computer programs. Let's see the fundamental reason why that simplicity makes them insecure in today's world. Remember that a cipher's permutation should look random in order to be secure. Of course, the best way to look random is to be random—that is, to select every

permutation randomly from the set of all permutations. And there are many permutations to choose from. In the case of the 26-letter English alphabet, there are approximately 2^{88} permutations:

$$26! = 403291461126605635584000000 = 2^{88}$$